Attention, Step by Step

A complete numerical walkthrough of one single-head self-attention pass.

Setup

We compute self-attention for the sentence

$$\text{“I like cats”}.$$

The sentence only gives names to the rows. The embeddings and projection matrices are deliberately chosen toy values, not learned model parameters.

Here $N = 3$ counts token rows, $d_{\text{model}} = 2$ is the embedding width, $d_k = 2$ is the query/key width used in dot products, and $d_v = 2$ is the value width mixed into the output.

$$x_I = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad x_{\text{like}} = \begin{bmatrix} 0 & 1 \end{bmatrix}, \qquad x_{\text{cats}} = \begin{bmatrix} 1 & 1 \end{bmatrix}.$$

Using row-token notation,

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

The fixed projection matrices are

$$W_Q = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad W_V = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix},$$

with $W_Q, W_K, W_V \in \mathbb{R}^{2\times 2}$.

The attention equation is

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.$$

1. Compute $Q = XW_Q$

Since W_Q is the identity matrix,

$$Q = XW_Q = XI_2 = X.$$

Therefore

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

2. Compute $K = XW_K$

$$K = XW_K.$$

$$k_I = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix} = \begin{bmatrix}(1)(1)+(0)(0) & (1)(1)+(0)(1)\end{bmatrix} = \begin{bmatrix}1 & 1\end{bmatrix},$$

$$k_{\text{like}} = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix} = \begin{bmatrix}(0)(1)+(1)(0) & (0)(1)+(1)(1)\end{bmatrix} = \begin{bmatrix}0 & 1\end{bmatrix},$$

$$k_{\text{cats}} = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1 & 1\\0 & 1\end{bmatrix} = \begin{bmatrix}(1)(1)+(1)(0) & (1)(1)+(1)(1)\end{bmatrix} = \begin{bmatrix}1 & 2\end{bmatrix}.$$

Thus

$$K = \begin{bmatrix}1 & 1\\0 & 1\\1 & 2\end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

3. Compute $V = XW_V$

$$V = XW_V.$$

$$v_I = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix} = \begin{bmatrix}(1)(1)+(0)(1) & (1)(0)+(0)(1)\end{bmatrix} = \begin{bmatrix}1 & 0\end{bmatrix},$$

$$v_{\text{like}} = \begin{bmatrix}0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix} = \begin{bmatrix}(0)(1)+(1)(1) & (0)(0)+(1)(1)\end{bmatrix} = \begin{bmatrix}1 & 1\end{bmatrix},$$

$$v_{\text{cats}} = \begin{bmatrix}1 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix} = \begin{bmatrix}(1)(1)+(1)(1) & (1)(0)+(1)(1)\end{bmatrix} = \begin{bmatrix}2 & 1\end{bmatrix}.$$

Thus

$$V = \begin{bmatrix}1 & 0\\1 & 1\\2 & 1\end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

4. Compute $K^\top$

Transposing K turns its columns into rows:

$$K^\top = \begin{bmatrix}1 & 0 & 1\\1 & 1 & 2\end{bmatrix} \in \mathbb{R}^{2\times 3}.$$

5. Compute the Raw Scores $S = QK^\top$

The raw attention score matrix is obtained by query-key dot products.

For $q_I = [1, 0]$:

$$q_I \cdot k_I = (1)(1) + (0)(1) = 1,$$
$$q_I \cdot k_{\text{like}} = (1)(0) + (0)(1) = 0,$$
$$q_I \cdot k_{\text{cats}} = (1)(1) + (0)(2) = 1.$$

For $q_{\text{like}} = [0, 1]$:

$$\begin{aligned} q_{\text{like}} \cdot k_I &= (0)(1) + (1)(1) = 1, \\ q_{\text{like}} \cdot k_{\text{like}} &= (0)(0) + (1)(1) = 1, \\ q_{\text{like}} \cdot k_{\text{cats}} &= (0)(1) + (1)(2) = 2. \end{aligned}$$

For $q_{\text{cats}} = [1, 1]$:

$$\begin{aligned} q_{\text{cats}} \cdot k_I &= (1)(1) + (1)(1) = 2, \\ q_{\text{cats}} \cdot k_{\text{like}} &= (1)(0) + (1)(1) = 1, \\ q_{\text{cats}} \cdot k_{\text{cats}} &= (1)(1) + (1)(2) = 3. \end{aligned}$$

Therefore

$$S = QK^\top = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

6. Compute the Scaled Scores

Here $d_k = 2$, so

$$\sqrt{d_k} = \sqrt{2} \approx 1.4142.$$

$$\frac{1}{\sqrt{2}} \approx 0.7071, \qquad \frac{2}{\sqrt{2}} \approx 1.4142, \qquad \frac{3}{\sqrt{2}} \approx 2.1213.$$

Thus

$$S_{\text{scaled}} = \frac{QK^\top}{\sqrt{d_k}} = \begin{bmatrix} 0.7071 & 0.0000 & 0.7071 \\ 0.7071 & 0.7071 & 1.4142 \\ 1.4142 & 0.7071 & 2.1213 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

Scaling prevents large dot products from pushing softmax too strongly toward saturated probabilities as the key/query dimension grows.

7. Compute $A = \text{softmax}(S_{\text{scaled}})$

Softmax is applied independently to each row. For a row vector $z = [z_1, z_2, z_3]$,

$$\text{softmax}(z)_j = \frac{e^{z_j}}{\sum_{\ell=1}^{3} e^{z_\ell}}, \qquad j = 1, 2, 3.$$

Equivalently,

$$\text{softmax}\big(\begin{bmatrix} z_1 & z_2 & z_3 \end{bmatrix}\big) = \frac{1}{e^{z_1} + e^{z_2} + e^{z_3}} \begin{bmatrix} e^{z_1} & e^{z_2} & e^{z_3} \end{bmatrix}.$$

First row

$$z = \begin{bmatrix} 0.7071 & 0.0000 & 0.7071 \end{bmatrix}.$$

$$e^{0.7071} \approx 2.0281, \qquad e^0 = 1.0000.$$

$$Z_I = 2e^{0.7071} + e^0 \approx 2(2.0281) + 1.0000 = 5.0562.$$

$$\frac{e^{0.7071}}{Z_I} \approx 0.4011, \qquad \frac{e^{0}}{Z_I} \approx 0.1978.$$

$$a_I = \begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \end{bmatrix}.$$

Second row

$$z = \begin{bmatrix} 0.7071 & 0.7071 & 1.4142 \end{bmatrix}.$$

$$e^{0.7071} \approx 2.0281, \qquad e^{1.4142} \approx 4.1132.$$

$$Z_{\text{like}} = 2e^{0.7071} + e^{1.4142} \approx 2(2.0281) + 4.1132 = 8.1694.$$

$$\frac{e^{0.7071}}{Z_{\text{like}}} \approx 0.2483, \qquad \frac{e^{1.4142}}{Z_{\text{like}}} \approx 0.5035.$$

$$a_{\text{like}} = \begin{bmatrix} 0.2483 & 0.2483 & 0.5035 \end{bmatrix}.$$

Third row

$$z = \begin{bmatrix} 1.4142 & 0.7071 & 2.1213 \end{bmatrix}.$$

$$\begin{aligned} e^{1.4142} &\approx 4.1132, \\ e^{0.7071} &\approx 2.0281, \\ e^{2.1213} &\approx 8.3420. \end{aligned}$$

$$Z_{\text{cats}} = e^{1.4142} + e^{0.7071} + e^{2.1213} \approx 4.1132 + 2.0281 + 8.3420 = 14.4833.$$

$$\text{softmax}(z) \approx \frac{1}{Z_{\text{cats}}} \begin{bmatrix} 4.1132 & 2.0281 & 8.3420 \end{bmatrix}.$$

$$a_{\text{cats}} = \begin{bmatrix} 0.2840 & 0.1400 & 0.5760 \end{bmatrix}.$$

The attention matrix is

$$A = \begin{bmatrix} 0.4011 & 0.1978 & 0.4011 \\ 0.2483 & 0.2483 & 0.5035 \\ 0.2840 & 0.1400 & 0.5760 \end{bmatrix} \in \mathbb{R}^{3\times 3}.$$

8. Verify the Row Sums of A

$$\begin{aligned} 0.4011 + 0.1978 + 0.4011 &\approx 1.0000, \\ 0.2483 + 0.2483 + 0.5035 &\approx 1.0001, \\ 0.2840 + 0.1400 + 0.5760 &\approx 1.0000. \end{aligned}$$

The tiny deviation in the second row is only rounding error.

9. Compute the Final Output $O = AV$

The final output mixes value vectors, not query vectors and not key vectors.

$$v_I = \begin{bmatrix} 1 & 0 \end{bmatrix}, \qquad v_{\text{like}} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad v_{\text{cats}} = \begin{bmatrix} 2 & 1 \end{bmatrix}.$$

For token I:

$$\begin{aligned} o_I &= 0.4011 v_I + 0.1978 v_{\text{like}} + 0.4011 v_{\text{cats}}, \\ o_{I,1} &= 0.4011(1) + 0.1978(1) + 0.4011(2) \approx 1.4011, \\ o_{I,2} &= 0.4011(0) + 0.1978(1) + 0.4011(1) \approx 0.5989. \end{aligned}$$

For token like:

$$\begin{aligned} o_{\text{like}} &= 0.2483 v_I + 0.2483 v_{\text{like}} + 0.5035 v_{\text{cats}}, \\ o_{\text{like},1} &= 0.2483(1) + 0.2483(1) + 0.5035(2) \approx 1.5035, \\ o_{\text{like},2} &= 0.2483(0) + 0.2483(1) + 0.5035(1) \approx 0.7517. \end{aligned}$$

For token cats:

$$\begin{aligned} o_{\text{cats}} &= 0.2840 v_I + 0.1400 v_{\text{like}} + 0.5760 v_{\text{cats}}, \\ o_{\text{cats},1} &= 0.2840(1) + 0.1400(1) + 0.5760(2) \approx 1.5760, \\ o_{\text{cats},2} &= 0.2840(0) + 0.1400(1) + 0.5760(1) \approx 0.7160. \end{aligned}$$

Therefore

$$O = \begin{bmatrix} 1.4011 & 0.5989 \\ 1.5035 & 0.7517 \\ 1.5760 & 0.7160 \end{bmatrix} \in \mathbb{R}^{3\times 2}.$$

10. Explain One Row of A

Each row of A describes how one token distributes its attention over all three value vectors.

For the *cats* row,

$$a_{\text{cats}} = \begin{bmatrix} 0.2840 & 0.1400 & 0.5760 \end{bmatrix},$$

the token *cats* assigns about 28.40% of its attention to I, about 14.00% to *like*, and about 57.60% to *cats*.

Final Question

For the token *cats*, the largest attention weight is on *cats* itself, because 0.5760 is the largest entry in the *cats* row of A.

Bonus Question

No. The token receiving the largest attention weight does not necessarily have the largest value vector. Recall that

$$A = \operatorname{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right), \qquad O = AV.$$

The attention weights, the percentages in matrix A, are determined entirely by the compatibility between the queries Q and keys K. The value vectors V are completely ignored during the softmax calculation; they are used only after the weights have been computed, when the final output O is formed.

Pipeline Summary

The computation can be read in three stages:

$$\boxed{\text{Tokens} \longrightarrow X \longrightarrow \begin{cases} Q = XW_Q, \\ K = XW_K, \\ V = XW_V. \end{cases}}$$

$$\boxed{(Q, K) \longrightarrow S = QK^\top \longrightarrow \hat{S} = \frac{S}{\sqrt{d_k}} \longrightarrow A = \operatorname{softmax}(\hat{S})}$$

$$\boxed{(A, V) \longrightarrow O = AV}$$

The full pipeline is

$$\boxed{\begin{array}{l} X \longrightarrow \begin{array}{l} Q = XW_Q \\ K = XW_K \\ V = XW_V \end{array} \longrightarrow S = QK^\top \longrightarrow \hat{S} = \dfrac{QK^\top}{\sqrt{d_k}} \\ \qquad\quad \longrightarrow A = \operatorname{softmax}(\hat{S}) \longrightarrow O = AV. \end{array}}$$

Collapsed into the single attention equation,

$$\boxed{O = \operatorname{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V.}$$

In summary,

$$\boxed{\begin{array}{c} X \longrightarrow (Q, K, V) \longrightarrow QK^\top \longrightarrow \dfrac{QK^\top}{\sqrt{d_k}} \\ \longrightarrow A \longrightarrow AV \longrightarrow O. \end{array}}$$

with

$$\boxed{\begin{array}{c} Q = XW_Q, \qquad K = XW_K, \qquad V = XW_V, \\ A = \operatorname{softmax}\left(\dfrac{QK^\top}{\sqrt{d_k}}\right). \end{array}}$$